



Moving averages

MOVING AVERAGES SHOW GENERAL TRENDS IN DATA OVER A CERTAIN PERIOD OF TIME.

What is a moving average?

When data is collected over a period of time, the values sometimes change, or fluctuate, noticeably. Moving averages, or averages over specific periods of time, smooth out the highs and lows of fluctuating data and instead show its general trend.

Showing moving averages on a line graph

Taking data from a table, a line graph of individual values over time can be plotted. The moving averages can also be calculated from the table data, and a line of moving averages plotted on the same graph.

The table below shows sales of ice cream over a two-year period, with each year divided into four quarters. The figures for each quarter show how many thousands of ice cream cones were sold.

	YEAR ONE				YEAR TWO			
Quarter	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
Sales (in thousands)	1.25	3.75	4.25	2.5	1.5	4.75	5.0	2.75

△ Table of data

These figures can be presented as a line graph, with sales shown on the y axis and time (measured in quarters of a year) shown on the x axis.

▷ Sales graph

The sales graph shows quarterly highs and lows (pink line), while a moving average (green line) shows the trend over the two-year period.

REAL WORLD

Seasonality

Seasonality is the name given to regular changes in a data series that follow a seasonal pattern. These seasonal fluctuations may be caused by the weather, or by annual holiday periods such as Christmas or Easter. For example, retail sales experience a predictable peak around the Christmas period and low during the summer vacation period.

▷ Ice cream sales

Sales of ice cream tend to follow a predictable seasonal pattern.



Calculating moving averages

From the figures in the table, an average for each period of four quarters can be calculated and a moving average on the graph plotted.

Average for quarters 1–4

Calculate the mean of the four figures for year one. Mark the answer on the graph at the midpoint of the quarters.

$$1.25 + 3.75 + 4.25 + 2.5 = 11.75$$

$$\frac{11.75}{4} = 2.94$$

sum of sales figures for quarters 1–4 mean value (rounded to 2 decimal places)

number of values

